

Assignment II

Mathematics for Computing

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Module Overview

1 Descriptive Statistics

2 Frequentist Statistics

Introduction to Descriptive Statistics

Definition

Descriptive statistics involves summarizing and organizing data so that it can be easily understood. Unlike inferential statistics, it seeks to describe the features of a specific data set without making broader generalizations.

- **Visualizing Data:** Using graphical representations like histograms to observe data distribution.
- **Measures of Central Tendency:** Capturing the center point of data (e.g., sample mean).
- **Measures of Dispersion:** Understanding data spread (e.g., sample variance).

Histograms and Graphical Representation

- **Histogram:** A precise graphical representation of the distribution of numerical data.
- It groups continuous data into underlying intervals called **bins**.

Key Characteristics:

- Area of each bar is proportional to the frequency of observations in the bin.
- Helps identify underlying distributions (skewness, modality, outliers).

Interpretation

A symmetric bell-shaped histogram suggests normal distribution, while a tail extending right indicates a positively skewed dataset.

Sample Mean and Variance

- Let $X = \{x_1, x_2, \dots, x_n\}$ be a sample of size n .

Sample Mean ($\bar{x}$)

Provides the central location of the sample data:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Sample Variance (s^2)

Measures the degree of spread or dispersion around the mean:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Note: The denominator uses $n - 1$ (Bessel's correction) to ensure an unbiased estimator.

Order Statistics

- **Definition:** Order statistics are the sample values arranged in ascending order.
- Let $X_1, X_2, \dots, X_n$ be random variables. The ordered sample values are denoted by:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

Common Order Statistics

- **Minimum:** $X_{(1)}$
- **Maximum:** $X_{(n)}$
- **Median:** The middle value, representing $X_{(\frac{n+1}{2})}$ for odd n .

Sample Covariance and Covariance Matrix

- **Sample Covariance:** Measures the linear relationship between two distinct numerical variables, X and Y .

$$q_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Sample Covariance Matrix (Σ)

For a multivariate dataset with p features, the covariance matrix organizes the variances and covariances into a symmetric $p \times p$ structure:

$$\Sigma = \begin{bmatrix} s_1^2 & q_{12} & \dots & q_{1p} \\ q_{21} & s_2^2 & \dots & q_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ q_{p1} & q_{p2} & \dots & s_p^2 \end{bmatrix}$$

Frequentist Statistics: Core Philosophy

- **The Paradigm:** Interprets probability as the limit of relative frequencies in a large number of independent trials.
- Parameters are treated as **fixed but unknown constants**, while data is treated as random.

Statistical Inference Goals

- ① **Estimation:** Approximating unknown population parameters using sample metrics.
- ② **Hypothesis Testing:** Deciding whether experimental evidence supports a particular assumption.

Sampling and Estimators

- **Sampling:** Drawing a representative subset from a larger statistical population to draw inferences.
- **Estimator:** A rule or formula (statistic) used to estimate an unknown population parameter θ . It is denoted as $\hat{\theta}$.

Point vs Interval Estimation

- **Point Estimate:** A single numerical value calculated from the sample data (e.g., using $\bar{x}$ to estimate population mean μ).
- **Interval Estimate:** A range of values within which the parameter is expected to fall.

Properties of Estimators: MSE and Consistency

Mean Square Error (MSE)

Measures the average squared difference between the estimator $\hat{\theta}$ and the parameter θ :

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

Consistency

An estimator is **consistent** if it converges in probability to the true parameter value as the sample size n approaches infinity:

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| < \epsilon) = 1 \quad \text{for any } \epsilon > 0$$

As you gather more data, a consistent estimator approaches the exact truth.

Confidence Intervals

- **Definition:** An interval computed from sample statistics that contains the true parameter value with a specified probability level $(1 - \alpha)$.

General Formulation

For a population mean μ with known standard deviation σ :

$$CI = \bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

Frequentist Interpretation

If the sampling experiment is repeated many times, $(1 - \alpha)\%$ of the generated confidence intervals will cover the true parameter. It does **not** mean the parameter has a probability of being inside that fixed specific interval.

Model Estimation

- Statistical models help formalize data-generation mechanisms into mathematical rules.

Parametric Model Estimation

- Assumes data originates from a known family of probability distributions indexed by a finite set of parameters (e.g., Normal Distribution defined entirely by μ and σ^2).
- Methods include **Maximum Likelihood Estimation (MLE)** and **Method of Moments**.

Non-Parametric Model Estimation

- Does not assume the structure of the distribution beforehand.
- Parameters can be infinite-dimensional or grow with sample size (e.g., Kernel Density Estimation, Empirical Distribution Functions).